

AES IntelliNet

7744F / 7788F

Complete Technician Guide

Programming, timing, zones, diagnostics, routing,
serial wiring, monitor decoding and commissioning

FIELD SCOPE

- AES 7744F and 7788F legacy fire subscribers
- AES 7041 handheld programmer and PC/Tera Term equivalents
- Verified US232R / DB9 breakout connection
- Based on AES Rev 6 installation and operation manuals

1. Scope, safety and how to use this guide

This manual is a field reference for AES IntelliNet 7744F and 7788F legacy subscriber units. It covers the AES 7041 handheld programmer key functions and their PC/Tera Term equivalents. The programming language is similar on both models, but the physical behavior of several zone inputs differs.

Life-safety warning

Before powering, resetting, programming, keying or testing an installed subscriber, notify the central station and place the account on test. A subscriber can transmit immediately after power-up or reset. Programming must conform to the approved system design, current code requirements, the authority having jurisdiction and the central station's AES network requirements.

What this guide includes

- Verified US232R and DB9-breakout wiring to radio connector J1.
- Complete handheld-to-PC command conversion.
- Every documented programming, monitor and diagnostic command for these models.
- Timer ranges, defaults and the practical effect of each timer.
- 7788F EOL-zone behavior and 7744F EOL/reversing-voltage differences.
- Status, self-test codes, monitor-packet fields, routing entries and signal-quality interpretation.
- A full commissioning, troubleshooting and closeout sequence.

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2. Serial connection and verified wiring

Tera Term configuration

Setting	Required value	Reason
Port	COM1	The FTDI US232R port used during the verified field test.
Speed	4800 baud	AES programmer-port data rate.
Data bits	8	Required serial word length.
Parity	None	No parity bit is used.
Stop bits	1	Required framing.
Flow control	None	AES explicitly requires RTS/CTS flow control OFF.
Local echo	Normally OFF	The subscriber echoes prompts and responses; local echo can duplicate characters.
Backspace	Destructive, if available	Erases the deleted character from the terminal display.

Verified J1-to-DB9 pinout

Connector orientation is critical: view the 6P6C modular plug at the copper-contact end with the contacts facing you. Read physical J1 pins from left to right. The J1 pin functions remain fixed even when the cable colors are reversed.

Physical J1 pin functions - authoritative for both cable versions

Physical J1 pin	Function	DB9 terminal
1	GND	Pin 5 GND
2	TXD	Pin 3 TXD
3	RXD	Pin 2 RXD
4	DTR	Pin 4 DTR
5	DSR	Pin 6 DSR
6	+12 V from radio	DO NOT CONNECT
-	RTS/CTS local jumper	Pin 7 RTS to Pin 8 CTS

2. Standard-color RJ cable wiring

Case A - verified cable order: BLUE, YELLOW, GREEN, RED, BLACK, WHITE

Copper contacts facing you, left to right:

PIN 1	PIN 2	PIN 3	PIN 4	PIN 5	PIN 6
BLUE	YELLOW	GREEN	RED	BLACK	WHITE
GND	TXD	RXD	DTR	DSR	+12 V

J1 pin	Wire color	DB9 terminal	Function
1	Blue	Pin 5 GND	Signal ground
2	Yellow	Pin 3 TXD	PC/adaptor transmit path
3	Green	Pin 2 RXD	PC/adaptor receive path
4	Red	Pin 4 DTR	Programmer control line
5	Black	Pin 6 DSR	Programmer status line
6	White	DO NOT CONNECT	+12 V supplied by the radio
-	-	Pin 7 RTS to Pin 8 CTS	Local jumper at breakout

2. Reversed-color RJ cable wiring

Use this case only when the plug at radio J1 is viewed at the copper contacts and the colors read WHITE, BLACK, RED, GREEN, YELLOW, BLUE from physical pin 1 through pin 6.

Case B - reversed cable order: WHITE, BLACK, RED, GREEN, YELLOW, BLUE

Copper contacts facing you, left to right:

PIN 1	PIN 2	PIN 3	PIN 4	PIN 5	PIN 6
WHITE	BLACK	RED	GREEN	YELLOW	BLUE
GND	TXD	RXD	DTR	DSR	+12 V

J1 pin	Wire color	DB9 terminal	Function
1	White	Pin 5 GND	Signal ground
2	Black	Pin 3 TXD	PC/adaptor transmit path
3	Red	Pin 2 RXD	PC/adaptor receive path
4	Green	Pin 4 DTR	Programmer control line
5	Yellow	Pin 6 DSR	Programmer status line
6	Blue	DO NOT CONNECT	+12 V supplied by the radio
-	-	Pin 7 RTS to Pin 8 CTS	Local jumper at breakout

J1 pin 6 is always the dangerous conductor

Physical J1 pin 6 is approximately +12 V from the radio. In Case A it is WHITE; in the reversed Case B it is BLUE. Insulate it. Connecting physical pin 6 to a DB9 or FTDI signal terminal can damage the adapter, computer or subscriber.

Do not trust color without verifying orientation

Identify the copper-contact face and physical pin numbers first. Before connecting an unfamiliar cable, use continuity testing and confirm the radio's +12 V conductor with a meter. Never energize the adapter while the pin-6 identity is uncertain.

FT_PROG polarity

Use standard FT232R polarity. Leave every FT_PROG signal-inversion option OFF. The working COM1 test produced readable SETUP UNIT and ENTER ID prompts with standard polarity.

Case A verified wiring diagram

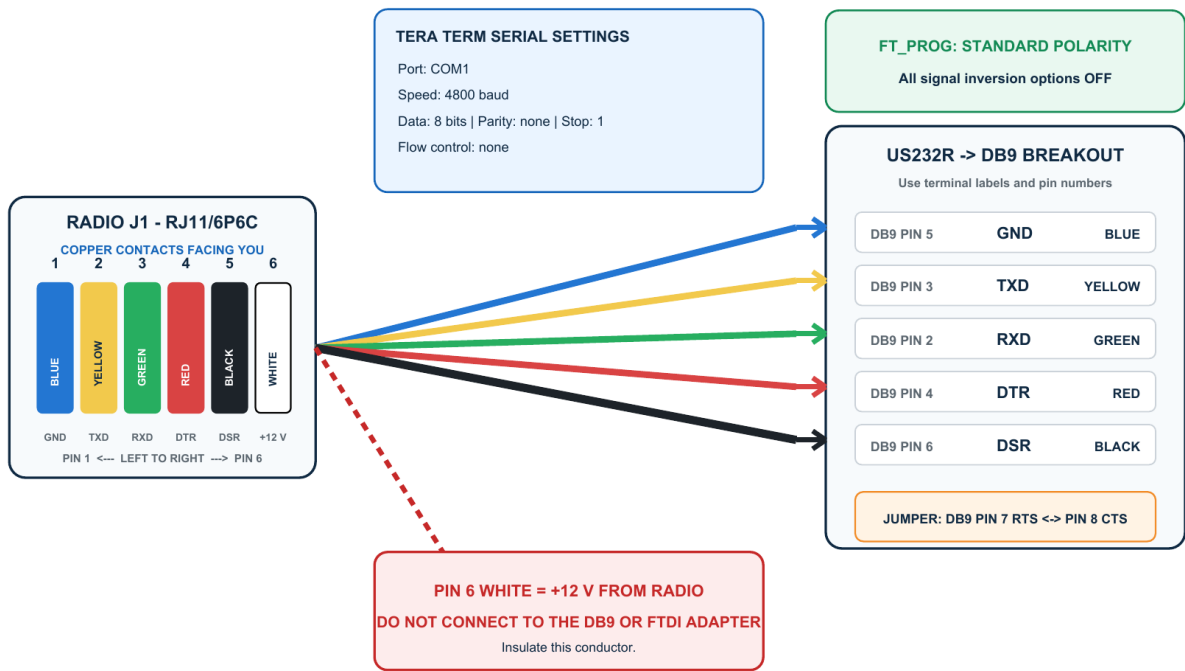
This image applies only when the copper-contact order is BLUE, YELLOW, GREEN, RED, BLACK, WHITE. For the reversed WHITE, BLACK, RED, GREEN, YELLOW, BLUE order, use the Case B table on the preceding page.

AES IntelliNet 7744F / 7788F

US232R / DB9 BREAKOUT WIRING

FINAL WORKING CONNECTION

6P6C modular plug shown at the copper-contact end, contacts facing you.



This exact wiring produced readable serial output from the radio during the COM1 test.

Verified field wiring | Page 1 of 2

If the terminal receives readable prompts but cannot send commands, recheck TXD/RXD, confirm flow control is None, and verify the pin 7-to-pin 8 jumper. If all text is unreadable, reconfirm 4800 8-N-1 and that FT_PROG inversion is off.

3. Complete command map

The handheld key name and the PC key are not always the same. For example, handheld Ctrl+F1 is a single lower-case f on a PC terminal. Letter commands execute as soon as the character is received. Press Enter only when the subscriber requests a value.

Handheld key	PC / Tera Term	Function
Ctrl+F1	f	Program subscriber ID and cipher
Ctrl+F2	g	Program check-in, AC delay and report spacing
Ctrl+F3	h	Program Fire/Trouble packets, zones and restorals
Ctrl+F4	i	Program repeating and fault-report suppression
Ctrl+F5	j	Reset configuration RAM; retains ID and cipher
Shift+F1	a	RX Monitor toggle
Shift+F2	b	TX Monitor toggle
Shift+F3	c	Monitor All toggle; RX Monitor must already be ON
Shift+F4	d	Local Status Check
Shift+F5	e	Key transmitter for approximately 5 seconds
F1	Ctrl+Q	Key equivalent; no standalone quick action documented
F2	Ctrl+R	Key equivalent; no standalone quick action documented
F3	Ctrl+S	Key equivalent; no standalone quick action documented
F4	Ctrl+T	Display routing table
F5	Ctrl+U	Send a text message to the central station
Ctrl+Z	Ctrl+Z	Display current zone status

Data-entry rules

- Use upper-case A-F when entering hexadecimal IDs or cipher values.
- At most programming prompts, pressing Enter without entering a new value preserves the old value.
- The subscriber normally displays OLD and NEW fields. Verify the old value before changing it.
- Programming functions normally time out after approximately 65 seconds.
- If Tera Term appears not to show what you type, remember that local echo is normally off; watch for the subscriber response.
- Use the physical Reset button after zone-programming changes and after completing monitor work.

F1 through F3

The AES manuals provide the PC equivalents Ctrl+Q, Ctrl+R and Ctrl+S but do not assign separate standalone field functions to handheld F1, F2 or F3 for the 7744F/7788F. Do not invent a function based on another AES model.

4. f - Program subscriber ID and cipher

```
SETUP UNIT-OLD: NEW  
ENTER ID#-1234:....  
CPHR CODE-XXXX:....  
OK
```

Subscriber ID

The ID is the subscriber's unique four-digit hexadecimal network address. Valid characters are 0-9 and upper-case A-F. Examples include 4923, 12AF and B804. Every subscriber on the AES network must have a unique ID.

- Entering a new value overwrites the stored ID.
- Pressing Enter at a blank ID prompt preserves the current ID.
- A reused ID can send signals under the wrong account and create serious dispatch or reporting problems.

System cipher

The cipher is a four-digit hexadecimal network security code assigned by the AES network administrator. It must match the cipher used by the central receiver or the IP-Link through which the subscriber communicates.

- The old cipher is intentionally displayed as XXXX.
- Pressing Enter without a new cipher preserves the stored cipher.
- A wrong cipher prevents normal network enrollment and communication.

Symptoms of an incorrect cipher or invalid network identity

- WA indicator continues flashing.
- The subscriber does not receive packet acknowledgements.
- Local status may show RT1:XXXX or LEVEL:255.
- The radio may transmit, but the unit never becomes a healthy network participant.

Local-only programming

ID and cipher can only be changed with a programmer or PC terminal attached directly to J1. Remote programming of these two values is not permitted. A RAM reset does not erase them.

5. g - Timing parameters

The Timing Parameters menu programs three independent values.

Parameter	Range / choices	Factory default	Primary purpose
Check-in	Up to 24 hours; 00:00 illegal	24:00	Periodic subscriber-alive supervision
AC report delay	00-60 minutes or RM	RM	Disperse AC fail/restore traffic
NTR report delay	0-330 seconds	10 seconds	Space the subscriber's own transmissions

Check-in interval

```
CHKIN TIME--OLD:NEW
ENTER HRS----24:..
ENTER MINS---00:..
```

A check-in is a periodic supervisory data message confirming that the subscriber is alive and participating in the network. It is not an alarm. Shorter intervals permit quicker recognition of a missing subscriber but increase RF and central-station traffic.

- A 24-hour interval is typical except in higher-security applications.
- 00 hours, 00 minutes is illegal.
- Do not program a total interval greater than 24 hours.
- The central automation missing-check-in threshold must agree with the programmed interval.

Manual range wording

The display documents hours as 00-24, while the explanatory text says 1-24 and separately states that 00:00 is illegal. This indicates zero hours can be used with a nonzero minute value on compatible firmware. Use only the interval approved for the network and installation.

AC report delay

```
AC RPT TIME-OLD:NEW [RM] or [00-60]
ENTER MINS---RM:..
```

This value is applied after the subscriber recognizes an AC failure or AC restore. The subscriber requires approximately 100 minutes of continuous AC loss to recognize a failure, and approximately 100 minutes of restored AC to recognize a restore.

Setting	Effect
RM	Random additional delay from 0 to 60 minutes; factory default
00	No additional delay after the approximately 100-minute detection period
01-60	Fixed additional delay in minutes

With RM, an AC event is normally reported approximately 100-160 minutes after the physical condition begins. Random delay prevents large numbers of subscribers from transmitting simultaneously during an area-wide outage.

5. Timing parameters - continued

NTR report delay

NTR RPT DLY-NNN:... [0-330 seconds]

This is the minimum spacing applied to the subscriber's own reports. The default and recommended setting is 10 seconds. It meters multiple events so the subscriber does not continuously occupy the network.

- Range: 0 to 330 seconds.
- Default: 10 seconds.
- Zero removes the spacing delay and is not recommended.
- Commercial fire and applicable UL burglary installations require no more than 20 seconds.
- If the previous delay has already expired, a newly generated event can begin transmission immediately.

Delay versus suppression

AC report delay determines when an AC event is transmitted. AC suppression in the i menu prevents the event from being transmitted at all. Do not confuse these settings.

Timing examples

Example	What happens
Check-in 24:00	The subscriber creates a check-in about once every 24 hours.
AC delay RM	After about 100 minutes of AC loss, an additional random 0-60 minute delay is selected.
AC delay 00	After about 100 minutes of AC loss, the AC-fail message is ready without an added delay.
NTR delay 10	The subscriber normally spaces its own queued reports by at least 10 seconds.
AC suppressed Y	No AC-fail report is sent, regardless of the AC-delay value.

6. h - Zone programming

Zone programming controls Fire/Trouble packet reporting, the electrical interpretation of every zone and whether each zone reports a return to normal.

Reset required after zone changes

After completing zone programming, press the physical Reset button. The reset initializes the inputs under their new configuration and clears former conditions. Without it, an input changed to Bypass or another type can continue to show a confusing old alarm or trouble state.

Fire/Trouble packet prompt

```
FIRE/  
TROUBLE PKT-N: . [Y/N]
```

Selection	Behavior
Y	Transmit zone-trouble messages. Required for applicable fire/UL installations. Disables normal remote programming.
N	Do not transmit zone-trouble messages. Not appropriate for required fire trouble reporting.

When Fire/Trouble packets are enabled, normal central-station remote programming is disabled. Completing f, g or h through a locally connected programmer can temporarily open an approximately 10-minute remote-programming window. The window begins after the local function is completed or after its 65-second timeout.

Zone type entry

```
ZONE BANK 0  
SET ZONE (BSF) B0  
OLD SSSSSSSS LOW>HI  
NEW .....
```

Enter exactly eight letters, in zone order from 1 through 8. Entering more or fewer than eight values causes the prompt to repeat.

Code	Name	EOL-zone behavior
B	Bypassed	Input is ignored; no normal zone alarm/trouble processing
S	Supervised	Normal at approximately 2.2 kOhm; alarm on open or short
F	Fire supervised	Normal at approximately 2.2 kOhm; trouble on open; alarm on short

Why S and F are different

Supervised S identifies both an open and short as an alarm-type abnormal condition. Fire-supervised F distinguishes a broken/open circuit as trouble and a short circuit as alarm. This distinction is normally required for fire-service inputs.

6. Zone programming - model differences

7788F

All eight 7788F zone inputs are optically isolated 2.2 kOhm EOL inputs. Every input can be configured as B, S or F.

7744F

The 7744F uses two different physical input types.

Zones	Hardware	S behavior	F behavior
1-4	2.2 kOhm EOL	Alarm on open or short	Trouble on open; alarm on short
5-8	Reversing-voltage	Alarm on polarity reversal or low/no voltage	Alarm on reversal; trouble on low/no voltage

Although both models accept the same eight-letter programming string, the physical meaning of zones 5-8 is different on a 7744F. Confirm the board model before diagnosing a zone from resistance or voltage readings.

Restoral programming

```
SET RESTORAL (XR) B0
OLD XXXXXXXX LOW>HI
NEW .....
```

Code	Behavior
R	Transmit a report when the zone returns to its programmed normal state.
X	Do not transmit a return-to-normal report. Factory default.

Restoral selection does not control whether the original alarm or trouble is sent. It controls only the later normal transition. For example, a short can create an alarm report and the return to 2.2 kOhm creates a restoral only if that zone is R.

Zone-programming defaults

Parameter	Factory default
Fire/Trouble packets	N - not transmitted
Zone type	S on all eight zones
Restoral	X on all eight zones - not transmitted

7. i - Operating modes

```
SET MODES--OLD:NEW
ENABLE RPTNG-Y: .
SPRSS ACFAIL-N: .
SPRSS CHRGFLT-N: .
SPRSS GNDFLT-N: .
```

Enable repeating

Value	Behavior
Y	Subscriber may receive and relay packets for other AES units. Factory default.
N	Subscriber does not serve as a packet repeater for other units.

Repeating is a core mesh-network function. Disabling it may not prevent the subscriber from reporting its own events, but it removes the unit as a possible route for neighboring subscribers and can weaken the surrounding network. Leave repeating enabled unless the AES network administrator specifically directs otherwise.

Suppress AC failure

Y prevents AC-failure reports from being transmitted. N permits normal AC-failure reporting and is the factory default. UL installations require N. Suppression does not change AC detection; it hides the report.

Suppress charger fault

Y prevents battery-charger fault reports from being transmitted. N permits reporting and is the factory default. A charger fault can indicate that the subscriber is unable to maintain its standby battery. UL installations require N.

Suppress ground fault

Y prevents ground-fault reports from being transmitted. N permits reporting and is the factory default. Suppression does not correct an unintended connection to chassis/earth ground. UL installations require N.

Recommended normal mode values

Repeating Y; suppress AC fail N; suppress charger fault N; suppress ground fault N. Any deviation should be documented and approved for the particular installation.

8. j - Reset RAM and the physical Reset button

Reset RAM command

j
Y
ENTER

The j command restores most programmable parameters to factory defaults and then starts a self-test. It preserves the subscriber ID and cipher.

Parameter	After RAM reset
ID and cipher	Preserved
Check-in	24 hours
AC report delay	RM - random 0-60 minutes
NTR report delay	10 seconds
Fire/Trouble packets	N
Zone types	S on all zones
Restorals	X - not reported
Repeating	Y - enabled
AC, charger and ground-fault suppression	N - reporting allowed

Use RAM reset for a previously used unit, an unknown configuration, suspected configuration corruption or a troubleshooting procedure specifically requiring defaults. Do not use it casually on a commissioned fire installation because the default zone and trouble-reporting values may not meet the approved design.

Physical Reset button

The board Reset button restarts the controller, runs the self-test, restarts network enrollment and turns monitor modes off. It does not erase normal programming.

Operation	Normal programming erased?	ID/cipher retained?	Monitor modes cleared?
Physical Reset	No	Yes	Yes
j RAM reset	Most settings	Yes	Yes

9. d - Local Status Check

```
SUB [revision] 7788F
ID#:4923 (C)YYYY AES
RT1:1234 LEVEL:002
STAT:000 NETCON:3
```

Model and revision

The first line identifies the subscriber model and firmware revision.

ID

The locally programmed four-digit subscriber ID.

RT1

RT1 is the first-choice next hop at the top of the subscriber's dynamic routing table.

RT1 display	Meaning
Another subscriber ID	That unit is the current preferred first hop.
IP-Link ID	The subscriber is likely communicating directly with that IP-Link.
0000	Direct central/receiver-related route, depending on network configuration.
XXXX	Subscriber is not currently on the network.

LEVEL

Value	Interpretation
1	Expected direct communication with the receiver/IP-Link.
2	Expected route through one other subscriber.
3	Expected route through two other subscribers.
255	Not on the network.

Lower levels are generally preferable, but a stable multi-hop path can be valid. The displayed level can be increased if the top route has weak signal quality or reports faults.

NETCON

NETCON is an internal Network Connectivity confidence rating based on available routes and their quality; it is not simply a received-signal-strength reading.

- 0 is best.
- 0 through 6 are generally usable under these manuals.
- 7 can indicate a problem with the subscriber or a unit in its route.
- Fire alarm installations require NETCON 0 through 5.

9. Status and self-test interpretation

STAT and SELFTEST error values

Status values are additive. A combined number can represent more than one condition.

Code	Meaning	Initial action
000	No self-test fault reported	Continue normal verification
001	Low battery	Check battery voltage, capacity and connections; reset
002	RAM checksum failure	Reset; consider RAM reset if directed
003	001 plus 002	Correct battery issue and investigate RAM
004 / 006	EEPROM failure	Reset; service may be required
008	Analog-to-digital converter failure	Reset; service may be required
080	Radio loopback-test failure	Reset and recheck; service if persistent
100	AC not present	Check transformer, branch circuit and AC wiring
101	AC absent plus low battery	Correct both power conditions
200	Battery charger failure	Inspect battery/charger; service may be required
300	100 plus 200	AC absent and charger fault; battery-only operation may explain it
400	Ground fault, where supported	Locate unwanted ground on zone/input wiring

Example: 084 equals loopback failure 080 plus EEPROM failure 004. Repeated internal failures after reset may require AES-authorized service.

Board LED interpretation

LED	Normal diagnostic meaning
ALM	Steady periodic blink indicates normal. Other patterns indicate low battery, zone abnormality or self-test failure.
WA	On means waiting for acknowledgement; steady blinking means not on network; off is normal.
TX	Illuminates while the radio transmitter is active.
RX	Illuminates when RF activity opens/uses the receiver. More than about 20 seconds steady can indicate interference.

ALM blink patterns

Pattern	Meaning
Steady single blink pattern	System OK
Short-short	Low battery
Short-long	Zone alarm or trouble
Short-short-long	Low battery plus zone alarm/trouble
Steady/no blink	Self-test failure other than low battery

10. Monitor and traffic diagnostics

a - RX Monitor

RX Monitor displays decoded network packets addressed to this subscriber. It proves that the subscriber is receiving and decoding traffic intended for it. It does not display every carrier or every packet exchanged by other subscribers.

b - TX Monitor

TX Monitor displays decoded protocol messages transmitted by this subscriber, including check-ins, alarms, troubles, text messages, retries and network-management packets.

c - Monitor All

Monitor All expands receive monitoring to decoded AES network traffic within RF range. RX Monitor must already be enabled.

```
a -> RX MONITOR ON
c -> MONITOR ALL ON
b -> TX MONITOR ON (optional, adds local transmissions)
```

Monitor All does not display every signal that opens squelch. The RX LED can illuminate for interference, another protocol or an undecodable signal without producing a terminal line.

Packet-line format

```
<1234 #1A-> 0000 1234 -> FF01 <D_CHKIN>>
```

Field	Meaning
First 1234	ORIG - subscriber that originally created the packet
1A	PN - packet sequence number from 00 through FF
0000	DEST - final destination; 0000 commonly represents the central receiver
Second 1234	FROM - unit currently transmitting the packet
FF01	TO - immediate next-hop recipient
D_CHKIN	Packet identifier - in this example, a data check-in

Common packet identifiers

Identifier	Meaning
D_CHKIN	Subscriber check-in data
P_ACK	Packet acknowledgement
P_RR	Receiver Ready broadcast used to help maintain routing information

10. Monitor diagnostics - practical use

What each monitor proves

Observation	What it proves	What it does not prove
RX Monitor line	A packet for this subscriber was decoded	That all traffic in the area is visible
TX Monitor line	The subscriber created/transmitted a protocol packet	That another unit acknowledged it
Monitor All line	A nearby AES packet was decoded	That the packet belongs to this account/network
RX LED only	RF energy opened/used the receiver	That valid AES data was decoded
TX LED only	The transmitter keyed	That a useful packet reached the central station

Why Monitor All can show ON but remain blank

- RX Monitor a was not enabled before Monitor All c.
- No AES packets are currently being transmitted within usable range.
- The antenna, coax or radio receiver path is defective.
- The received signal is interference or a different protocol.
- Traffic is too weak or corrupted to decode.
- The radio frequency/channel does not match the surrounding AES network.

Monitor cautions

- a, b and c are toggles; pressing the same letter again turns that mode off.
- Incoming text messages cannot be received while any monitor function is active.
- Turn all monitor functions off when finished, or press the physical Reset button.

Recommended monitor sequence

Reset the subscriber, wait for enrollment, press a, then c, then b if local transmit traffic is also required. Record several complete packet/acknowledgement pairs before judging the route.

11. e - Key transmitter

KEYING TX..
TIMEOUT

The transmitter is keyed for approximately five seconds and then shuts off automatically. Press Enter during the test to abort early; OK is displayed after an early abort. TIMEOUT at the end of the full interval is normal.

Appropriate uses

- Measure transmitter output power with an RF wattmeter.
- Measure forward and reflected power or antenna-system SWR.
- Check coax, connectors, antenna tuning and transmit current draw.
- Confirm that the radio and TX indicator key.

What the key test does not prove

- Correct packet modulation.
- Correct subscriber ID or cipher.
- Network enrollment.
- Reception or acknowledgement by another subscriber.
- Delivery of an event to the central station.

Because this function primarily keys the carrier for RF measurements, it may not create a normal decoded TX Monitor line.

Ctrl+Z - Display Zone Status

Zx12,Z1-8:0100-1000

The hexadecimal value is a bit pattern representing abnormal zone-input conditions. The character string gives the state of zones 1 through 8 from left to right, with a hyphen between zones 4 and 5.

Character	Meaning
0	Zone is normal under its current programming
1	Faulted, alarm or otherwise not normal
T	Trouble

Use this command immediately after zone programming and reset. An unexpected 1 or T can indicate a wrong zone type, missing 2.2 kOhm EOL resistor, open/short circuit, reversed polarity, low/no voltage on a 7744F reversing input, or stale state because the unit was not reset after programming.

11. Ctrl+T - Display Routing Table

```
4.5678,L:02,N:5,Q:83
3.7888,L:01,N:5,Q:03
2.BB12,L:00,N:0,Q:03
1.AA11,L:00,N:0,Q:03
```

The PC terminal can display as many as eight route entries. The handheld's four-line screen normally shows only the highest-priority entries. Entries are displayed in reverse visual order so route preference 1 remains visible on the handheld.

Field	Meaning
1. through 8.	Route preference; 1 is the current first choice
ID	Subscriber or IP-Link available as a next hop
L	That neighbor's Link Layer / Level
N	That neighbor's NETCON
Q	Received carrier/data quality

Signal quality Q

Q has two meaningful characters.

Example	Interpretation
03	Carrier detect was active and all three repeated data copies were decoded; best
02	Carrier detect active; two of three copies decoded
01	Carrier detect active; only one copy decoded
83	Carrier detect was not solid, but all three copies decoded
82 / 81	Weak or marginal carrier and only two/one copies decoded

The first character is 0 when carrier detect was on and 8 when it was not reliably on. The second character is the number of the three repeated data copies that were decoded. Therefore 03 is best and 81 is among the poorest usable observations.

Route changes

The table is dynamic. If the preferred route repeatedly fails to return a packet acknowledgement, the subscriber can remove that entry from preference 1 and promote another route. The manuals describe removal after seven failed attempts.

12. Ctrl+U - Text messages

ENTER MSG:

[Enter up to 200 ASCII characters, then press ENTER]

A text message can be sent from the subscriber to the central station. The subscriber exits text-entry mode if no data is entered for approximately 65 seconds.

- Maximum message length: 200 ASCII characters.
- Press Enter to send.
- The subscriber cannot perform normal transmit or receive operation while text-entry mode is active.
- Incoming central-station text is displayed on the connected programmer/terminal and normally produces a beep.
- Incoming text cannot be received while any monitor mode is active.

Text message versus event test

A successful text message demonstrates useful two-way network communication, but it is not a substitute for testing every required alarm, trouble and restoral event with the central station.

13. Troubleshooting

Readable prompts but commands do not work

- Verify flow control is None.
- Verify DB9 pin 7 is jumpered to pin 8 at the breakout.
- Recheck yellow J1 pin 2 to DB9 pin 3 TXD.
- Use lower-case a through j, not the handheld key combination on the PC.
- Ensure no other program has COM1 open.

Unreadable or garbled output

- Confirm 4800 baud, 8 data bits, no parity and 1 stop bit.
- Confirm FT_PROG inversion is OFF for every FT232R signal.
- Confirm blue J1 pin 1 reaches DB9 pin 5 ground.
- Confirm green J1 pin 3 reaches DB9 pin 2 RXD.
- Do not connect physical J1 pin 6: it is white in Case A and blue in reversed Case B.

Subscriber transmits but does not enroll

- Verify the four-digit ID is valid and unique.
- Verify the cipher matches the network.
- Inspect antenna, coax, connectors and frequency/channel.
- Watch WA: continuous blinking means the subscriber is not on the network.
- Use d; RT1:XXXX or LEVEL:255 confirms no usable network enrollment.

13. Troubleshooting - continued

Monitor All is blank

- Enable a before c.
- Confirm nearby network activity exists.
- Observe the RX LED; no RX activity suggests no usable RF energy is reaching the receiver.
- A steady RX LED for more than about 20 seconds can indicate interference.
- Keying with e is a carrier test and may not create decoded traffic.

Zone reads alarm or trouble unexpectedly

- Confirm the correct model: 7744F zones 5-8 are reversing-voltage inputs.
- Confirm the zone's B/S/F programming.
- Measure the EOL resistance at the subscriber terminals; account for connected-panel loading.
- Check for open, short, reversed polarity or low/no voltage.
- Press physical Reset after zone changes, then check Ctrl+Z again.

Acknowledgement delay

An acknowledgement delay occurs when a transmitted data packet does not receive its packet acknowledgement within the programmed acknowledgement-delay time. The manuals identify a 60-second default acknowledgement delay. Investigate route quality, packet retries, WA behavior, the routing table and surrounding network health.

When a persistent self-test code remains

- Press physical Reset and record the complete code.
- Correct power, battery, charger or ground-fault conditions first.
- Use RAM reset only when appropriate and after recording programming.
- A persistent EEPROM, ADC or loopback failure may require AES-authorized service.

13. Commissioning and closeout checklist

Step	Technician action	Acceptable result
1	Notify central station; place account on test	Testing authorized and documented
2	Verify wiring, antenna, battery and AC	No unsafe or reversed connections
3	Physical Reset	SELFTEST-PASS or understood power-only code
4	Run d Local Status	Correct model/ID; STAT acceptable; RT1 valid; LEVEL not 255; NETCON compliant
5	Run Ctrl+T Routing Table	Multiple healthy routes when available; Q values understood
6	Run Ctrl+Z Zone Status	Every zone agrees with field condition
7	Review f identity	Unique correct ID and correct cipher
8	Review g timers	Check-in/delays match approved requirements
9	Review h zones	Fire/Trouble, B/S/F and restorals match drawings
10	Review i modes	Repeating and suppression settings approved
11	Use a/c/b diagnostics as needed	Packets and acknowledgements observed; monitors then turned off
12	Test every required alarm/trouble/restoral	Correct central-station account, zone and event received
13	Restore normal conditions and reset	No unexplained alarm, trouble or monitor state
14	Remove account from test	Central station confirms normal
15	Document final values	Service record updated and retained

Do not close the job on RF indication alone

TX LEDs, Monitor output and a good route are useful diagnostics, but final acceptance requires confirmation that the monitoring system received and correctly interpreted every required signal.

Suggested programming record

Field	Recorded value
Model / firmware	_____
Subscriber ID	_____
Cipher verified by	_____
Check-in	_____ hours _____ minutes
AC report delay	_____
NTR report delay	_____
Fire/Trouble packets	Y / N
Zone types 1-8	_____
Restorals 1-8	_____
Repeating	Y / N
AC / charger / ground suppression	_____
RT1 / Level / STAT / NETCON	_____
Central station test confirmation	_____

14. Technician quick reference

PC key	Function	Critical note
f	ID and cipher	Upper-case hex; blank Enter preserves value
g	Timers	24 h / RM / 10 s defaults
h	Zones	Reset after changes
i	Modes	Normal: repeating Y; suppression N
j	RAM reset	Erases most settings; keeps ID/cipher
a	RX Monitor	Packets addressed to this subscriber
b	TX Monitor	Packets transmitted by this subscriber
c	Monitor All	Requires a already ON
d	Status	Check RT1, LEVEL, STAT, NETCON
e	Key transmitter	Carrier/power test; not network proof
Ctrl+T	Routing table	Review preference, L, N and Q
Ctrl+U	Text message	Up to 200 characters
Ctrl+Z	Zone status	0 normal, 1 abnormal, T trouble

Normal serial and FTDI setup

COM1 | 4800 baud | 8 data bits | no parity | 1 stop bit
 Flow control: None | FT_PROG inversion: all OFF
 DB9 pin 7 RTS jumpered to pin 8 CTS

Normal monitor sequence

Physical Reset
 a RX Monitor ON
 c Monitor All ON
 b TX Monitor ON, if needed
 Toggle a/b/c OFF or reset when finished

Normal key configuration defaults

Check-in 24:00 | AC delay RM | NTR delay 10 seconds
 Trouble packets N | Zones SSSSSSSS | Restorals XXXXXXXX
 Repeating Y | AC/charger/ground suppression N

Official sources and document notes

AES 7744F Installation and Operation Manual

Document 40-7744, Revision 6, August 30, 2016

<https://aes-corp.com/wp-content/uploads/2020/07/7744-Install-Manual.pdf>

AES 7788F Installation and Operation Manual

Document 40-7788, Revision 6, August 30, 2016

<https://aes-corp.com/wp-content/uploads/2020/07/40-7788-Rev-6.pdf>

FTDI FT232R datasheet

https://ftdichip.com/wp-content/uploads/2020/08/DS_FT232R.pdf

Known manual typographical issues

The AES Rev 6 manuals contain a few apparent editing inconsistencies, including a 7744F zone-reset paragraph that names the 7788F, and a testing checklist that can confuse Ctrl+T routing-table and Ctrl+Z zone-status usage. The detailed command sections are used in this guide: Ctrl+T displays the routing table and Ctrl+Z displays zone status.

Guide status

This technician guide consolidates the information required for field programming and diagnostics. It does not replace the full AES installation manuals, approved drawings, central-station procedures, listing limitations or applicable code.

Verified wiring in this guide was tested using an FTDI US232R/FT232R USB-to-serial adapter, a DB9 breakout board and COM1. A readable SETUP UNIT / ENTER ID prompt was obtained with standard FTDI signal polarity.

Technician	Date	Site / account	Subscriber ID